10 HEAT STRUCTURE THERMAL PROPERTIES

These cards are used in NEW or RESTART problems. These cards are required if cards 1CCCGXNN, Heat Structure Input cards, Section 9 are entered. These data, if present, are processed and stored even if no cards 1CCCGXNN are entered.

The subfield MMM is the composition number, and the cards with this subfield describe the thermal properties of composition MMM. The composition numbers entered on cards 1CCCG201-1CCCG299 correspond to this subfield. A set of cards 201MMMNN must be entered for each composition number used, but MMM need not be consecutive. During RESTART, thermal property may be deleted, new compositions may be added, or data may be modified by entering new data for an existing composition.

10.1 Composition Type and Data Format, 201MMM00

This card is required.

W1(A) Material type. Thermal properties for four materials are stored within the program: carbon steel (C-STEEL), stainless steel (S-STEEL), uranium dioxide (UO2), and zirconium (ZR). These properties are selected by entering the name in parentheses for this word. At present, the data are primarily to demonstrate capability. The user should check whether the data are satisfactory.

If a user-supplied table or function is to be used, enter TBL/FCTN for this word.

The word DELETE may be entered in RESTART problems to delete a composition.

The next two words are required only if TBL/FCTN is entered for W1.

W2(I) Thermal conductivity format flag or gap mole fraction flag.

Enter 1 if a table containing temperature and thermal conductivity is to be entered.

Enter 2 if functions are to be entered.

Enter 3 if the gap conductance model is used, and thus a table containing gas component names and mole fractions is to be entered.

W3(I) Volumetric heat capacity flag.

Enter -1 if a table containing only volumetric heat capacities is to be entered and the temperature values are identical to the thermal conductivity table.

Enter 1 if a table containing temperature and volumetric heat capacity is to be entered.

Enter 2 if functions are to be entered.

10.2 Constant Thermal Conductivity, 201MMM01-49

One of these cards is required if W1 of card 201MMM00 contains TBL/FCTN, W2 of contains a 1, and the thermal conductivity is constant for all temperatures.

W1(R) Constant thermal conductivity (W/m•K, Btu/s•ft•^{o}F).

10.3 Temperature-Thermal Conductivity, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 of contains a 1. In this case, the thermal conductivity varies with temperature, so the user has to enter pairs of temperatures and thermal conductivities. More than one pair can be entered on a single card. The total number of pairs is limited to 100. The temperatures must be in increasing order. The first and last temperatures must bracket the expected temperatures during the calculation. A code failure will occur if the calculated material temperature is outside the bracketed range.

W1(R) Temperature (K, ^{o}F).

W2(R) Thermal conductivity (W/m•K, Btu/s•ft•^{o}F).

10.4 Thermal Conductivity Function, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 contains a 2. Sets of nine quantities are entered, each set containing one function and its range of application. The function is

k = A0 + A1*TX + A2*TX**2 + A3*TX**3 + A4*TX**4+ A5*TX**(-1)

where TX = T - C, T is the temperature argument, and C is a constant. Each function has a lower and upper limit of application. The first function entered must be for the lowest temperature range. The lower limit of each following function must equal the upper bound of the previous function.

W1(R) Lower limit temperature (K, ^{o}F).

W2(R) Upper limit temperature (K, ^{o}F).

W3(R) A0 (W/m•K, Btu/s•ft•^{o}F).

W4(R) A1 (W/m•K^2, Btu/s•ft•$^oF^2$).

W5(R) A2 (W/m•K^3, Btu/s•ft•$^oF^3$).

W6(R) A3 (W/m•K^4, Btu/s•ft•$^oF^4$).

W7(R) A4 (W/m•K^5, Btu/s•ft•$^oF^5$).

W8(R) A5 (W/m, Btu/s•ft).

W9(R) C (K, ^oF).

10.5 Gap Mole Fraction, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 contains a 3, Enter pairs of gas component names and mole fractions. More than one pair can be entered on a single card. One to 7 pairs of gas names and their mole fractions can be entered. The gas component names that may be entered are HELIUM, ARGON, KRYPTON, XENON, NITROGEN, HYDROGEN, and OXYGEN. No particular order of the pairs is required. Do not enter any gas component with a zero mole fraction. Normalization of the total mole fraction to one is performed if the sum of the mole fractions entered is not one. The table of gas composition data is applicable to any gap and is required if card 1CCCG001 is present.

W1(R) Gas name. Possible names include HELIUM, ARGON, KRYPTON, XENON, NITROGEN, HYDROGEN, and OXYGEN.

W2(R) Mole fraction.

10.6 Volumetric Heat Capacity, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a -1. The card numbers need not be consecutive. More than one value can be entered on a single card. Since this format assumes the temperatures are identical to the values in the temperature-thermal conductivity table, only the volumetric heat capacities need be entered. The number of entries must be the same as the number of pairs in the temperature-thermal conductivity table, which is limited to 100.

W1(R) Volumetric heat capacity (J/m^3K, Btu/ft^3 ^oF). This is ρC_p, where ρ is density (kg/m^3, lb/ft^3) and C_p is specific heat capacity (J/kg·K, Btu/lb·^oF).

10.7 Constant Volumetric Heat Capacity, 201MMM51-99

One of these cards is required if W1 of card 201MMM00 contains TBL/FCTN, W2 of contains a 1, and the volumetric heat capacity is constant for all temperatures.

W1(R) Constant volumetric heat capacity (J/m^3K, Btu/ft^3 ^oF). This is ρC_p, where ρ is density (kg/m^3, lb/ft^3) and C_p is specific heat capacity (J/kg·K, Btu/lb·^oF).

10.8 Temperature-Volumetric Heat Capacity, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a 1. The card numbers need not be consecutive. More than one pair can be entered on a single card. Since the volumetric heat capacity varies with temperature, enter pairs of temperatures and volumetric heat capacities. More than one pair can be entered on a single card. The total number of pairs is limited to 100. The temperatures must be in increasing order. The first and last temperatures must bracket the expected temperatures during the calculation. A code failure will occur if the calculated material temperature is outside the bracketed range.

W1(R) Temperature (K, ^oF).

W2(R) Volumetric heat capacity (J/m^3K, $Btu/ft^3\ ^oF$). This is ρC_p, where ρ is density (kg/m^3, lb/ft^3) and C_p is specific heat capacity ($J/kg{\cdot}K$, $Btu/lb{\cdot}^oF$).

10.9 Volumetric Heat Capacity Function, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a 2. The card numbers need not be consecutive. Sets of nine quantities are entered, each set containing one function and its range of application. More than one set can be entered on a single card. The function is

$$\rho c_r = A0 + A1*TX + A2*TX**2 + A3*TX**3 + A4*TX**4 + A5*TX**(-1),$$

where TX = T - C, and T is the temperature argument. Each function has a lower and upper limit of application. The first function entered must be for the lowest temperature range. The lower limit of each following function must equal the upper bound of the previous function.

W1(R) Lower limit temperature (K, ^oF).

W2(R) Upper limit temperature (K, ^oF).

W3(R) A0 ($J/m^3\ K$, $Btu/ft^3\ ^oF$).

W4(R) A1 ($J/m^3\ K^2$, $Btu/ft^3\ ^oF^2$).

W5(R) A2 ($J/m^3\ K^3$, $Btu/ft^3\ ^oF^3$).

W6(R) A3 ($J/m^3\ K^4$, $Btu/ft^3\ ^oF^4$).

W7(R) A4 ($J/m^3\ K^5$, $Btu/ft^3\ ^oF^5$).

W8(R) A5 (J/m^{3}·, Btu/ft^{3}).

W9(R) C (K, ^{o}F).